

Microeconomics 2024

Part A

1. In an exchange economy Andrea and Bee each have Cobb-Douglas utility functions over goods x and y . Andrea's consumption bundle is $\{x_a, y_a\}$ and Bee's is $\{x_b, y_b\}$. The utility functions are, for Andrea first and Bee second,

$$u_a(x_a, y_a) = 2(\ln x_a + \ln y_a)$$

$$u_b(x_b, y_b) = \ln x_b + \ln y_b.$$

Andrea has an endowment of 18 units of good x and 6 units of good y . Bee's endowment is 6 units of good x and 18 units of good y .

- (a) Let the price of good y be 1, so the relative price of good x is p . Show that the competitive equilibrium relative price of x is $p = 1$, and that in the equilibrium allocation Andrea consumes 12 units of each good and Bee consumes 12 units of each good. State the conditions for an allocation to be Pareto efficient, and show that the allocation $(12, 12; 12, 12)$ satisfies them.
- (b) A social planner can transfer good y between the two agents to alter the initial endowments. Good x cannot be transferred. Explain why such a transfer does not change the equilibrium relative price.
- (c) If the planner is utilitarian, and so maximizes the sum of the utilities, how much of good y should be transferred (starting from the initial endowments), and who should receive the transfer?

[25%]¹

¹Split proposal for marking: 55% for part (a), 15% for part (b), 30% for part (c).

2. Alicia and Bob have quasi-linear utility functions over the quantity of a public good, q , and the money spent in a private good, x_a for Alicia and x_b for Bob. Alicia's utility function is

$$U_a(q, x_a) = 2q - q^2 + x_a.$$

Bob's utility function is

$$U_b(q, x_b) = q - \frac{1}{2}q^2 + x_b.$$

The cost of buying a unit of the public good is c , which satisfies $0 < c < 1$. Assume that Alicia and Bob have incomes, m_a and m_b respectively, that are large enough that each will consume a positive amount of the private good, so $x_a > 0$ and $x_b > 0$.

- (a) Suppose that Alicia and Bob independently choose how much of the public good to buy. How much would they buy if the other person buys nothing? Show that the Nash equilibrium with private provision of the public good has Alicia buying $1 - \frac{c}{2}$ and Bob buying zero of the public good. Explain.
- (b) What, in general, is the Samuelson condition for public goods provision? How much of the public good is provided when the Samuelson condition holds in this economy?
- (c) Professor Lindahl knows that Alicia and Bob always tell the truth. He suggests to the government that (i) each agent should be asked to report their marginal utility function for the public good, (ii) the Samuelson rule should determine the quantity of the public good, and (iii) each agent should pay a price per unit of the public good equal to their marginal utility at the quantity provided. (Equivalently each agent could be asked to report their demand function for the public good, with the government choosing the individual prices so that the demands are equal and the sum of the prices equals c .)

Discuss the reasoning behind Professor Lindahl's proposal, and whether it is likely to be a good scheme in practice.

[25%]²

²Split proposal for marking: 50% for part (a), 20% for part (b), 30% for part (c).

3. Row and Column play the following 2x2 strategic-form game. (Row picks the row, either T or B; Column picks the column, either L or R.) Row's payoff is written first in each cell. Both players know the value of x , to be specified below.

	L	R
T	2, 1	1, x
B	3, 1	0, 2

- (a) Assume $0 < x < 1$. Find all Nash equilibria of the game in which Row and Column choose simultaneously. Compute the players' equilibrium payoffs. How do the players' equilibrium strategies and payoffs depend on x ? Provide an explanation (intuition) for your findings.
- (b) Assume $0 < x < 1$. Explain whether Row or Column would weakly prefer to move first (committing to an action that is publicly observed) rather than choosing simultaneously.
- (c) Is there any two-player strategic-form game with a unique Nash equilibrium (in pure or mixed strategies), in which both players are strictly worse off moving first, provided their move is observed by the other player?
- (d) Assume instead $x = 1$. Find all Nash equilibria of the game with simultaneous moves.

[25%]³

³Split proposal for marking: 40% for part (a), 25% for part (b), 15% for part (c), 20% for part (d).

4. Ava is an expected utility maximiser with monetary utility $u(x)$, where u is strictly increasing. Consider the lotteries $L = [1 - p, p; 0, y]$ and $L' = [1 - q, q; 0, z]$, where the probabilities p and q belong to $(0, 1)$ and the monetary prizes satisfy $z > y > 0$.

- (a) Assume first that $q \geq p$. Does any of the lotteries first-order stochastically dominate the other?
- (b) Show that if Ava prefers L to L' , it must be that $p > q$. If we assume $u(0) = 0$ and $u(z) = 1$, what are the possible values of $u(y)$ in this case?
- (c) Prove formally that an expected utility maximiser with monetary utility $v(x) = \alpha + \beta u(x)$, where $\beta > 0$, has the same risk preferences as Ava does. Use this to defend the idea that the assumptions $u(0) = 0$ and $u(z) = 1$ above are harmless.

[25%]⁴

⁴Split proposal for marking: 30% for part (a), 30% for part (b), 40% for part (c).

Part B

5. Should microeconomic policy be guided by the principle that a policy should be implemented if the winners can compensate the losers and still be better off? If so, explain why. If not, how should microeconomic policy be determined?
6. For *either* emissions of greenhouse gases *or* traffic congestion in cities, discuss the relative merits of the following methods of addressing the externalities: (i) establishing and enforcing property rights, (ii) Pigouvian taxes, (iii) tradeable permits (cap-and-trade) schemes and (iv) direct regulation (which limits by law emissions from each source, or limits when a vehicle may enter a city).
7. Discuss, in the context of one or more suitable models of imperfect competition, under what conditions firms would prefer to merge rather than collude (and vice versa), and whether (or when) a regulator may prefer one outcome or the other (merger versus collusion).
8. A sordid club offers the following opportunity: you can win an obscenely huge prize or you can be dispossessed of everything you own. The club proudly claims that they can convince any expected utility maximiser to play by personalising the probability of winning in the game. The club also argues that, given a probability and a prize, any expected utility maximiser who wants to play the game would also want to put their name on a list from which one name would be randomly chosen to participate in the game. Using these two ideas, state formally the properties of continuity and independence and prove formally that expected utility satisfies them. Given a probability and a prize, what type of expected utility maximisers would you expect to play such a game?
9. “When information is symmetric, adverse selection and moral hazard are not relevant in the economic situation at hand; when information is asymmetric, adverse selection and moral hazard are equally relevant to understand the economic situation at hand.” Discuss.

10. A risk neutral firm considers hiring an agent for an important project. The project may end up in high (H), medium (M), or low (L) success, resulting in profits of 400 pounds, 280 pounds and 100 pounds, respectively. The probability of each type of success depends on whether the agent puts in high (h) or low (l) effort, as follows:

- If effort is high ($e_h = 1$), the corresponding probabilities are $\frac{1}{2}, 0$ and $\frac{1}{2}$.
- If effort is low ($e_l = 0$), the corresponding probabilities are $\frac{1}{4}, \frac{1}{4}$ and $\frac{1}{2}$.

The candidate is an expected utility maximizer with utility function $u(w, e) = \sqrt{100 + w} - e$. The candidate has a reservation utility of 11 and effort is not observable. The firm will be using contracts of the form (w_H, w_M, w_L) , contingent on the level of success of the project.

- (a) Discuss very briefly why the absence of observability leads to a moral hazard problem. If the firm wants to induce low effort, which contract will be offered? What would be the profits of the firm?

[15%]

- (b) Suppose that the firm wants to induce high effort. Write down the conditions that are needed for the agent to accept the contract and put in high effort.

[15%]

- (c) Suppose that the firm wants to induce high effort and can use any wages in $[-100, +\infty)$. Explain why setting a negative wage of $w_M = -100$ would be a good idea for the firm. Given this, why is a good idea to offer a contract in which $w_H = w_L$? What is then the optimal contract and the profits of the firm?

[20%]

- (d) Suppose that the firm wants to induce high effort but wages cannot be negative by law. Argue that the firm will set $w_M = 0$. Which are the optimal values of w_H and w_L for the firm in this situation?

[20%]

- (e) Comparing the profits of inducing both levels of effort, determine which contract will be offered by the firm if wages cannot be negative by law. Which is the agency cost?

[15%]

- (f) Suppose that the agent is allowed to give up the right to non-negative wages. Explain how this could generate a Pareto superior situation in which both the firm and the agent are strictly better off.

[15%]

Solution to Question 1.

- (a) The value of A's endowment is $18p + 6$ and the value of B's endowment is $6p + 18$. Since both preferences are Cobb-Douglas with equal coefficient, it can be argued that individuals will spend half of their income in each good.⁵ This leads to individual demands equal to

$$\begin{aligned} x_a(p) &= \frac{\frac{18p+6}{2}}{p} = 9 + \frac{3}{p} & y_a &= \frac{\frac{18p+6}{2}}{1} = 9p + 3, \\ x_b(p) &= \frac{\frac{6p+18}{2}}{p} = 3 + \frac{9}{p} & y_b &= \frac{\frac{6p+18}{2}}{1} = 3p + 9. \end{aligned}$$

By Walras' Law we can use market clearing in one market to determine the equilibrium price:⁶

$$x_a(p) + x_b(p) = 9 + \frac{3}{p} + 3 + \frac{9}{p} = 24,$$

so $p = 1$ as required.

Putting $p = 1$ into all four demand functions confirms that the consumption of each good for both individuals is 12.

The competitive equilibrium allocation is Pareto efficient as:⁷ (i) all of the goods are consumed, and (ii) the marginal rates of substitution are equal at the allocation:

$$\begin{aligned} MRS_a &= -\frac{\frac{2}{x_a}}{\frac{2}{y_a}} = -\frac{y_a}{x_a} = -\frac{12}{12} = -1, \\ MRS_b &= -\frac{\frac{1}{x_b}}{\frac{1}{y_b}} = -\frac{y_b}{x_b} = -\frac{12}{12} = -1. \end{aligned}$$

- (b) Denote the transfer from B to A as t . The resulting demand functions for good x are

$$\begin{aligned} x_a(p, t) &= \frac{1}{2} \frac{18p + 6 + t}{p} \\ x_b(p, t) &= \frac{1}{2} \frac{6p + 18 - t}{p}. \end{aligned}$$

Adding these we see that aggregate demand for x does not depend on t , so the equilibrium price remains $p = 1$. The key here is that the demand functions have the same functional form and are linear in income: the marginal propensities to consume x are equal. Demands only differ in the income that each agent has so a transfer of purchasing power makes no difference to aggregate demand.

- (c) By using all resources, the utilitarian objective can be written as

$$2(\ln x_a + \ln y_a) + \ln x_b + \ln y_b = 2 \ln x_a + \ln(24 - x_a) + 2 \ln y_a + \ln(24 - y_a).$$

⁵Candidates may naturally use the Lagrangian in their analysis. Also, candidates may clarify that the constant 2 plays no role for demand along their answer, but these details are not formally required for full marks.

⁶Candidates may obviously solve both markets.

⁷It would be good, but is not required, to mention that the First Fundamental Theorem of Welfare Economics applies. But if the only answer is that the First Fundamental Theorem of Welfare Economics applies that is not enough for a high mark (unless the First Theorem is itself proved).

Candidates may now solve directly for x_a and y_a by obtaining the first-order conditions of this concave problem.⁸ Condition $\frac{2}{x_a} - \frac{1}{24-x_a} = 0$ leads to $x_a = 16$, and $y_a = 16$ follows from the other condition. Overall, A is twice as good at converting goods into utility as B, and consumes two-thirds of the goods.

To obtain the appropriate transfer set

$$x_a(1, t) = \frac{1}{2} \frac{18 + 6 + t}{1} = 16$$

so $t = 8$. This goes from B to A.

⁸Notice that candidates may use also the following type of argument in finding the utilitarian solution: since this must be Pareto efficient, equality of the MRS's and the resource constraints imply

$$\frac{y_a}{x_a} = \frac{24 - y_a}{24 - x_a},$$

so $24x_a = 24y_a$ and $x_a = y_a$. This idea may have been discussed previously along their answers for parts (a) and (b). In such a case, the utilitarian objective becomes the one-variable function $4 \ln x_a + 2 \ln (24 - x_a)$, with same responses after that.

Solution to Question 2.

- (a) We can obtain the best responses as follows.⁹ A assumes that B will choose q_b . A chooses q_a to maximize

$$2(q_a + q_b) - (q_a + q_b)^2 + (m_a - cq_a)$$

with Kuhn-Tucker conditions:

$$2 - 2(q_a + q_b) - c \leq 0; q_a \geq 0; (2 - 2(q_a + q_b) - c)q_a = 0.$$

Whenever q_b is low enough, the solution is simply $q_a = 1 - \frac{c}{2} - q_b$, and zero otherwise.¹⁰

Similarly, B assumes that A will choose q_a . B chooses q_b to maximize

$$q_a + q_b - \frac{1}{2}(q_a + q_b)^2 + (m_b - cq_b)$$

with Kuhn-Tucker conditions

$$1 - (q_a + q_b) - c \leq 0; q_b \geq 0; (1 - (q_a + q_b) - c)q_b = 0.$$

Whenever q_a is low enough, the solution is simply $q_b = 1 - c - q_a$, and zero otherwise.

If $q_b = 0$, it must be $q_a = 1 - \frac{c}{2}$. If $q_a = 0$, it must be $q_b = 1 - c$. With the given values, $q_a = 1 - \frac{c}{2}$ and $q_b = 0$, all the Kuhn-Tucker conditions are satisfied with:

$$2 - 2(q_a + q_b) - c = 0; \quad q_a > 0.$$

$$1 - (q_a + q_b) - c < 0; \quad q_b = 0.$$

Intuitively A chooses her privately optimal quantity and B's marginal utility from additional q is strictly below c so B does not want to buy any of the public good. Each is optimizing given the choice of the other. A clear diagrammatic treatment of this would work.

- (b) The Samuelson condition is the condition for Pareto efficient allocation of the public good: the sum of the marginal rates of substitution between the public and private good should equal the marginal rate of transformation. With quasi-linear utilities and constant marginal cost, as here, the equivalent condition is that the sum of the marginal utilities equals the marginal cost.

Here this is

$$2(1 - q) + 1 - q = 3(1 - q) = c$$

so $q^* = 1 - \frac{c}{3}$, which exceeds the total provision in (a).

⁹Candidates may solve first under the assumption that the other person is not buying anything of the public good, as requested. From there, simple analysis helps them to obtain the Nash equilibrium. Since this question mostly corresponds to the public goods part of the paper, candidates should not be penalised for not computing best response functions formally, as long as their analysis is correct.

¹⁰Although not relevant, notice that for high values of q_b , individual a is willing to destroy part of the public good provided. The same applies to the other individual.

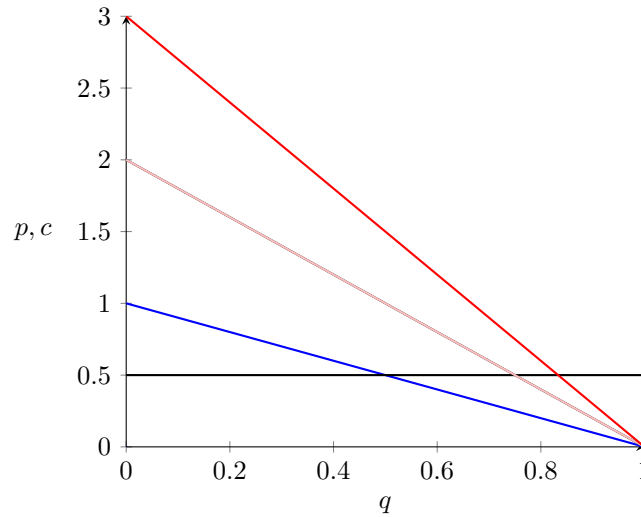


Figure 1: Lying and Lindahl

- (c) This is the Lindahl scheme. It achieves two objectives, with truth telling: (i) the Samuelson rule is applied so the provision is optimal and (ii) the total amount of revenue collected, by construction, equals the total cost of provision. A will face a price of $\frac{2c}{3}$ and B's price is $\frac{c}{3}$. The alternative way to implement the scheme has A and B submitting demand functions, $q_a(p_a) = 1 - \frac{p_a}{2}$ and $q_b(p_b) = 1 - p_b$, and then the government chooses p_a and p_b such that $q_a(p_a) = q_b(p_b)$, so $\frac{p_a}{2} = p_b$, and $p_a + p_b = c$. Full marks for this initial response would require either a very good technical presentation (that includes the computation of prices) or a very good conceptual presentation of the key aspects of the Lindahl pricing mechanism (that includes the economic intuitions on the pricing to individuals)

For the second part of the response, the main problem is the assumption of truth-telling. Each agent will typically have an incentive to lie. If B instead says that their marginal utility is zero, while A tells the truth, the outcome will be the same as in part (b). B's utility will be higher than with the truth (the gain from not paying anything for the public good is larger than the loss from the smaller provision of the good). Not required or expected to prove this, though a diagram from the lecture may be reproduced showing this. In the diagram $c = 0.5$ and the pink rectangle is the benefit B gets by not paying anything for the public good. Much smaller (not drawn) is the loss of utility as the quantity of the public good falls from 0.833 to 0.75. This is the area under the blue line (B's marginal utility) between 0.75 and 0.833.

Solution to Question 3.

- (a) There is no pure strategy Nash equilibrium for any $x \in (0, 1)$ because Row's unique best response to L is B (by $3 > 2$), Column's unique best response to B is R (by $2 > 1$), Row's unique best response to R is T (by $1 > 0$), Column's best response to T is L (by $1 > x$). Neither player is indifferent against the other's pure strategy, hence we must look for a fully-mixed equilibrium.

Suppose Row plays $pT + (1 - p)B$ with $p \in (0, 1)$. Then Column's indifference requires $1 = px + (1 - p)2$, equivalently $p = 1/(2 - x)$. Column's payoff is 1: it is constant in x because Column's payoff from L is always 1 and Column must be indifferent between L and R in when mixing. The weight Row puts on T is decreasing in x because Row is keeping Column indifferent, and as x increases T becomes more attractive to Column.

Suppose Column plays $qL + (1 - q)R$ with $q \in (0, 1)$. Row's indifference requires $2q + 1 - q = 3q$, equivalently $q = 0.5$. Row's payoff is 1.5. Neither Column's mixing weights nor Row's payoff depend on x because x is Column's payoff, which is irrelevant for keeping Row indifferent.

- (b) Row strictly prefers to move first rather than simultaneously: if she chooses T then Column responds with L, yielding a payoff of 2, which is strictly more than Row's mixed-equilibrium payoff. Intuition for why she prefers to commit: she strictly prefers Column's action L to Column's action R, and she can induce L by committing to T.¹¹

Column is indifferent between moving first or simultaneously. If he moves first then he'll choose L (inducing B, getting a payoff of 1) rather than R (inducing T, getting a payoff of $x < 1$). His payoff is the same as in the mixed equilibrium. He only weakly prefers Row's B to Row's T, so he ends up being indifferent.

- (c) The answer is yes. Examples are matching pennies or rock-paper-scissor.
- (d) (T,R) is now pure Nash. Column is indifferent between L and R against T, hence there is also a range of mixed Nash equilibria: $(T, qL + (1 - q)R)$ such that Row weakly prefers T to B, $2q + 1 - q \geq 3q$ equivalently $q \leq 0.5$. There is no Nash in which Row puts positive probability on B: in that case Column's strict best reply is R, to which Row's strict best reply is T, contradiction.

¹¹Candidates may babble about strategic substitutes / complements, and points should be deducted for this.

Solution to Question 4.

- (a) Candidates should formally construct the CDF of L and L' . The one corresponding to L is

$$\begin{aligned} F_L(x) &= 0 \text{ if } x < 0 \\ F_L(x) &= 1 - p \text{ if } x \in [0, y) \\ F_L(x) &= 1 \text{ otherwise.} \end{aligned}$$

The CDF of L' is

$$\begin{aligned} F_{L'}(x) &= 0 \text{ if } x < 0 \\ F_{L'}(x) &= 1 - q \text{ if } x \in [0, z) \\ F_{L'}(x) &= 1 \text{ otherwise.} \end{aligned}$$

Under the assumption that $q \geq p$, lottery the CDF of L' is always below that of L , and hence lottery L' first-order stochastically dominates lottery L . Candidates may naturally draw these CDFs and argue graphically.

- (b) Candidates may or may not use part (a). If they use it, notice that whenever $q \geq p$, L' first-order stochastically dominates L and any expected utility maximiser would then prefer L' . Moreover, this preference would be strict because $z > y$ and $q > 0$. Hence, if Ava prefers L to L' , it must be $p > q$. Candidates not using part (a) may argue by using the utility functional and comparing the expected utility of the lotteries: whenever $q \geq p$, the expected utility of L' is strictly greater than that of L and hence, it must be $p > q$. With the normalisation proposed, we can simply write $EU(L) \geq EU(L')$ as $pu(y) \geq q1$, i.e., $u(y) \geq \frac{q}{p}$. Since $u(y) < u(z)$, we learnt $u(y) \in [\frac{q}{p}, 1]$
- (c) Candidates can show how the ranking of any two lotteries is unaffected by an affine transformation of the monetary utilities. Full marks require to show that the ranking of all lotteries coincides, not only that of L and L' . Partial marks should be awarded for verification of the property for pair L, L' alone.

$$\begin{aligned} \sum_i p_i u(x_i) &\geq \sum_i p'_i u(x'_i) \\ \alpha + \beta[\sum_i p_i u(x_i)] &\geq \alpha + \beta[\sum_i p'_i u(x'_i)] \\ \sum_i p_i [\alpha + \beta u(x_i)] &\geq \sum_i p'_i [\alpha + \beta u(x'_i)] \\ \sum_i p_i v(x_i) &\geq \sum_i p'_i v(x'_i) \end{aligned}$$

We can simply choose the values of α and β for which the utility at 0 is zero and the utility at z is one. If u fails to have this property, just consider the values of α and β for which $v(0) = \alpha + \beta u(0) = 0$ and $v(z) = \alpha + \beta u(z) = 1$ and work with this normalisation.

Solution to Question 5. This is the compensation criterion, of Kaldor and Hicks, that aims to deal with the incompleteness of the Pareto criterion and purports only to use ordinal utility. The Pareto criterion says do a policy if there are winners and no losers, i.e. if there is a Pareto improvement, but these are very rarely available and it says nothing about what to do when there are winners and losers. So the compensation criteria were invented to try to skirt around this problem: the key is that the compensation does not have to be given (if there is compensation then the Pareto criterion applies). Some more advanced students, perhaps those who've done Public Economics or Environmental Economics, might note the Scitovsky paradox, that there can be cases where a policy satisfies the compensation criterion but so does reversing the policy. A good answer does not require discussion of this paradox. More important is to note that (i) the compensation criterion underlies the practice of cost-benefit analysis (everything is measured in money) and (ii) it presumes that moving a dollar from one person to another is neutral. This is hard to believe in a world with unequal income distribution: would a policy that takes \$100 from a very poor person and gives Elon Musk \$101 be good?

This suggests that we need to define a social welfare function. Of course there is then the problem that there is no agreement about what the social welfare function should be. Sometimes this doesn't matter: e.g. a utilitarian and a Rawlsian would each want to have equal utilities if the utility possibility frontier is strictly concave and symmetric (utility functions are the same). Social welfare functions presume that we can measure and compare utilities. Typically a social welfare function will be Paretian, so its maximization will require that the allocation is Pareto efficient (so there are no Pareto improvements possible). But note that a move from a Pareto inefficient allocation to an efficient one will not necessarily raise social welfare.

A very good answer will cover the Pareto criterion (do a policy if it leads to a Pareto improvement) and its limitations (incomplete, doesn't say what to do when someone is worse off), and the compensation criterion as a solution to the incompleteness problem. If compensation is given then the Pareto criterion applies: if compensation is not given, which is the idea, then the criterion smuggles in value judgements while pretending not to. Essentially it says that \$1 is equally valuable to everyone. Note, as covered in the lectures, that the compensation criterion is the basis of cost-benefit analysis. If the criterion is rejected, then need to define a social welfare function (which has its own problems, no agreement on its form other than that it should be Paretian). Some may say that policy should lead to a Pareto efficient outcome, this is a necessary condition for social welfare maximization but not sufficient, and there are many Pareto efficient allocations, and some moves to Pareto efficient allocations may be undesirable from a social welfare point of view. No need to cover Scitovsky for a first class answer: not done in core Micro.

Solution to Question 6. For both externality problems candidates aren't expected to have a lot of knowledge of the specifics, but they should understand how to evaluate the policy solutions using straightforward theory.

Greenhouse gases. (i) Property rights solution unworkable: too many polluters and victims, and most have not yet been born. Coasian solutions only work with small numbers, low transactions costs, full information about costs and benefits. (ii) Pigouvian taxes could work, and would generate revenue that could be used to offset distorting taxes such as income taxes. From a global perspective the tax should equal the marginal social cost of carbon (or equivalent gasses) and should be the same across the world: unlikely to get agreement on that. (iii) Cap-and-trade already used in EU, UK, parts of North America. Ideally permits should be tradeable across the world, with a single permit price that will equal the marginal social cost of carbon if the total number of permits is optimal. Candidates might discuss the Weitzman "prices-v-quantities" argument that a permit scheme is better if the marginal damage curve is relatively steep and a Pigouvian tax is better if marginal damage is relatively flat (compared to the marginal benefit of emissions). (iv) Direct regulation is often used: limiting technologies (no gas boilers, internal combustion engines), but may be inefficient because marginal abatement costs may differ across sources.

Traffic congestion. Focus here on the common resources problem. Environmental pollution is a smaller issue, and will not be a problem if all vehicles are electric and electricity is produced by renewables: the congestion problem will remain. (i) Property rights solution does not seem workable: would need bargaining between the drivers of different vehicles. (ii) Pigouvian tax: ideally pay per mile driven at different rates depending on how much congestion there is. Singapore's scheme is close to this. Problem: although this could in theory achieve efficiency no driver benefits: the gains all come from the tax revenue. In practice most road pricing schemes are cruder: pay £15 and you can drive in central London all day. (iii) Permits: potentially workable if the set of people entitled to permits can be determined (what about visitors, people coming from other countries?). As with road pricing ideally there should be different total numbers of permits at different times of the day. Using the Weitzman analysis it is likely that permits are better than taxes: the marginal damage in terms of time costs rises very steeply. (iv) Direct regulation: e.g. Athens allows cars to come in on different days of the week, enforced by number-plate recognition. Potential for inefficiency: what if I need to get to the hospital quickly to visit a dying relative and it's not the day for my car? But it is reasonably practical.

Solution to Question 7. In a Cournot game (covered in depth in lectures), e.g., with zero cost and linear demand, merger is not profitable unless a duopoly merges to monopoly. A good solution derives this technically and explains the intuition (the merged firm produces less than the two firms did prior to the merger; the merger benefits the other firms, not the merged entity). A merger can be profitable for the merging firms if it leads to cost savings.

A merger can be socially beneficial if the cost savings are sufficiently large. Williamson tradeoff: cost savings versus increase in market power. A good solution would explain this in the context of a model (at least a figure). A merger could have negative effects for society if it makes collusion easier (fewer firms).

If collusion can be sustained in an industry then the colluding firms obviously gain. If prices are kept high as a result then firms outside the collusive ring may gain as well. Consumers would typically lose (higher prices, lower production), the net effect is likely negative for society. Collusion may also impede technological progress or entry in an industry.

Solution to Question 8. Candidates will likely start presenting the lottery faced by individuals, $[p, 1 - p; w + H, 0]$, where p is the probability of the huge prize H , and w is the original wealth of the individual. Candidates may highlight that, according to the question, $w \ll H$.

The first part obviously corresponds to the property of continuity. In the version presented in lectures (see, e.g., Osborne and Rubinstein), we can create a lottery combining the best and worst prizes to match any other lottery. In the club case, this means to combine $w + H$ and 0 to match the current wealth of the individual (for her risk preferences). Candidates may very briefly argue that this may or may not be true, depending on the risk preferences of the individual. If they correspond to expected utility, then yes. Otherwise, not necessarily.

The second part obviously corresponds to the property of independence. In the version presented in lectures (see, e.g., Osborne and Rubinstein), the ranking of two lotteries is preserved if they are combined with a third lottery (with same probability). In the club case, the individual is asked to choose between current wealth, w , or the lottery $L = [p, 1 - p; w + H, 0]$. By writing your name on a list, the individual must choose between the lottery $[1; w] = [\frac{1}{N}, \frac{N-1}{N}; w, w]$ and the lottery $[\frac{1}{N}, \frac{N-1}{N}; L, w]$. Candidates may argue that the conclusion may or may not be true, depending on the risk preferences of the individual. If they correspond to expected utility, then yes. Otherwise, not necessarily.

The third part requires explicit formal analysis, if not provided before. Proofs that EU satisfy both properties are immediate and have been provided in lectures. They can be discussed in terms of the formal properties or in relation to the club versions, and that should not be penalised at all. For the club case, since $u(w+H) > u(w) > u(0)$, continuity of the functional $pu(W+H) + (1-p)u(0)$ shows that continuity is satisfied. For independence, notice that $\frac{1}{N}u(w) + \frac{N-1}{N}u(w) \geq \frac{1}{N}EU(L) + \frac{N-1}{N}u(w)$ if and only if $u(w) \geq EU(L)$, as required.

For the last part, candidates may naturally discuss the notions of risk aversion and more risk aversion, with some brief presentation of its measurement and the Arrow-Pratt theorem. They should argue that, for a given level of initial wealth, if a person participates, a less risk averse person will also participate (or argue negatively, on the no-participation). Some discussion of how wealth affects choices is needed. The usual *wealthier individuals may be less risk averse* type of argument with CARA/CRRA is correct but this should be read with caution because changes in wealth change the lottery. Moreover, it has been said that the prize H is huge, making payout $w + H$ relatively similar for everybody (the other being the same, 0) and wealth is thus secondary. The analysis of risk aversion for same wealth is thus more relevant here.

Solution to Question 9. Candidates are expected to argue that the first part is correct. They should probably define adverse selection and moral hazard, distinguishing clearly how information may create these two types of problems: understanding who is who in the market to determine optimal transactions and guaranteeing optimal actions of the contracted individual. Some examples may be expected. Candidates may simply argue that symmetric information rules out these two problems by definition (which, obviously, would require a good definition). A short discussion on efficient markets with symmetric information is also possible and useful in this part.

Candidates are expected to argue that the second part is incorrect. They may naturally explain how adverse selection and moral hazard create inefficiencies. A discussion of how adverse selection affects beliefs of the uninformed party, and how market failure can occur is relevant. A discussion of how moral hazard affects the type of contracts offered to risk averse individuals (agency cost included), and the inefficiencies that such contracts may provoke is relevant here. Up to this point, candidates can use different examples (e.g., lemons and principal-agent), and not necessarily a setting with both of them present (e.g., insurance). Fundamentally, candidates are expected to discuss that a specific setting may have one problem, the other, or both problems (and, crucially, in different amount/intensity/relevance). Candidates are expected to argue that the distinction is important because understanding which of the two phenomena is more relevant may help to understand the problematic better and how to eliminate the asymmetry and/or find solutions depends on this. Some discussion for each of the mechanisms in each of the two cases is expected.

Solution to Question 10.

- (a) In the absence of observability, the firm has no information on the effort chosen by the agent, key after contracting. This type of information asymmetry corresponds to the moral hazard type and candidates should briefly expand on this.

In order to induce low effort, given that the worker is risk averse, the cheapest way for the firm to do so is to pay a constant wage that covers exactly the reservation utility. That is,

$$\sqrt{100 + w} = 11 \text{ or equivalently } w = 21.$$

The profits of the firm would be

$$\frac{1}{4}400 + \frac{1}{4}280 + \frac{1}{2}100 - 21 = 220 - 21 = 199.$$

- (b) The individual rationality condition and the incentive compatibility constraint read:

$$\begin{aligned} \frac{1}{2}\sqrt{100 + w_H} + \frac{1}{2}\sqrt{100 + w_L} - 1 &\geq 11 \\ \frac{1}{2}\sqrt{100 + w_H} + \frac{1}{2}\sqrt{100 + w_L} - 1 &\geq \frac{1}{4}\sqrt{100 + w_H} + \frac{1}{4}\sqrt{100 + w_M} + \frac{1}{2}\sqrt{100 + w_L} \end{aligned}$$

- (c) The middle profit cannot appear if the worker puts in effort. Thus, negative wages with such middle profit do not affect the IR condition. For the IC constraint, these negative wages are undoubtedly better for the firm because they alleviate the constraints imposed by moral hazard. When $w_M = -100$, the IC reduces to

$$\begin{aligned} \frac{1}{2}\sqrt{100 + w_H} - 1 &\geq \frac{1}{4}\sqrt{100 + w_H} \text{ i.e.,} \\ \frac{1}{4}\sqrt{100 + w_H} &\geq 1. \end{aligned}$$

The IC constraint is thus trivially satisfied if the IR is satisfied and the moral hazard problem completely disappears. Since only individual rationality is relevant, the firm does not need to separate the wages (otherwise, will suffer the agency cost). The optimal contract must then satisfy

$$\frac{1}{2}\sqrt{100 + w} + \frac{1}{2}\sqrt{100 + w} - 1 = 11.$$

Thus, it must be $w_H = w_L = 44$. Profits are $\frac{1}{2}400 + \frac{1}{2}100 - 44 = 206$.

- (d) For the same reason discussed in (c), the firm wants w_M to be the smallest possible value, i.e., zero. Now, the incentive compatibility condition is no longer useless and must be considered. Notice that it now reads

$$\frac{1}{2}\sqrt{100 + w_H} - 1 \geq \frac{1}{4}\sqrt{100 + w_H} + \frac{1}{4}\sqrt{100} \text{ i.e.,}$$

$$\frac{1}{4}\sqrt{100 + w_H} \geq 3.5$$

This leads to $w_H = 96$ and, using the IR constraint, $w_L = 0$. The expected profit is $\frac{1}{2}(400 - 96) + \frac{1}{2}100 = 202$.

- (e) Since $202 > 199$, the firm prefers to induce high effort. Since the observability case would produce profits of 206 (this can be directly connected to part (c) or briefly analysed), the agency cost is equal to 4 units.
- (f) There is margin for a Pareto improvement strictly preferred by both parts. If wages are non-negative, there is an agency cost and the firm has some loss with respect to the optimal contract. Moreover, notice that in all cases considered so far, the agent is not obtaining any utility above the reservation value. For instance, contracts of the form $(w_H, w_M, w_L) = (44 + x, -100, 44 + x)$, with $x \in (0, 4)$ are efficient and strictly improve both sides.